

While building an advanced RAG system for a consulting client, I hit a retrieval quality problem where dense vector search alone was missing structurally important documents that keyword overlap would have caught. I implemented a hybrid retrieval layer combining pgVector semantic search with BM25 sparse retrieval, added a cross-encoder re-ranking step to score candidate chunks before passing context to the LLM, and introduced query decomposition to handle multi-part questions that were collapsing into a single ambiguous embedding. On the safety side, I built prompt injection detection at the input layer, enforced strict data isolation so proprietary client documents never mixed with public context, and added output validation guardrails to flag low-confidence or hallucinated responses before they reached the user. The system went from failing on roughly 30% of complex queries to handling them reliably, and the safety layer caught two injection attempts during internal testing.